Activity: Distance Sensor

Purpose: Joysticks are a great way to add some control to your projects. This circuit will walk you through the basics of using a joystick and how joysticks can be used to control objects. In this lesson, a joystick will be used to control two servo motors.

Materials:

Joystick, 2 servos, 12 jumper wires, Arduino Uno, breadboard

New Components:

Joystick

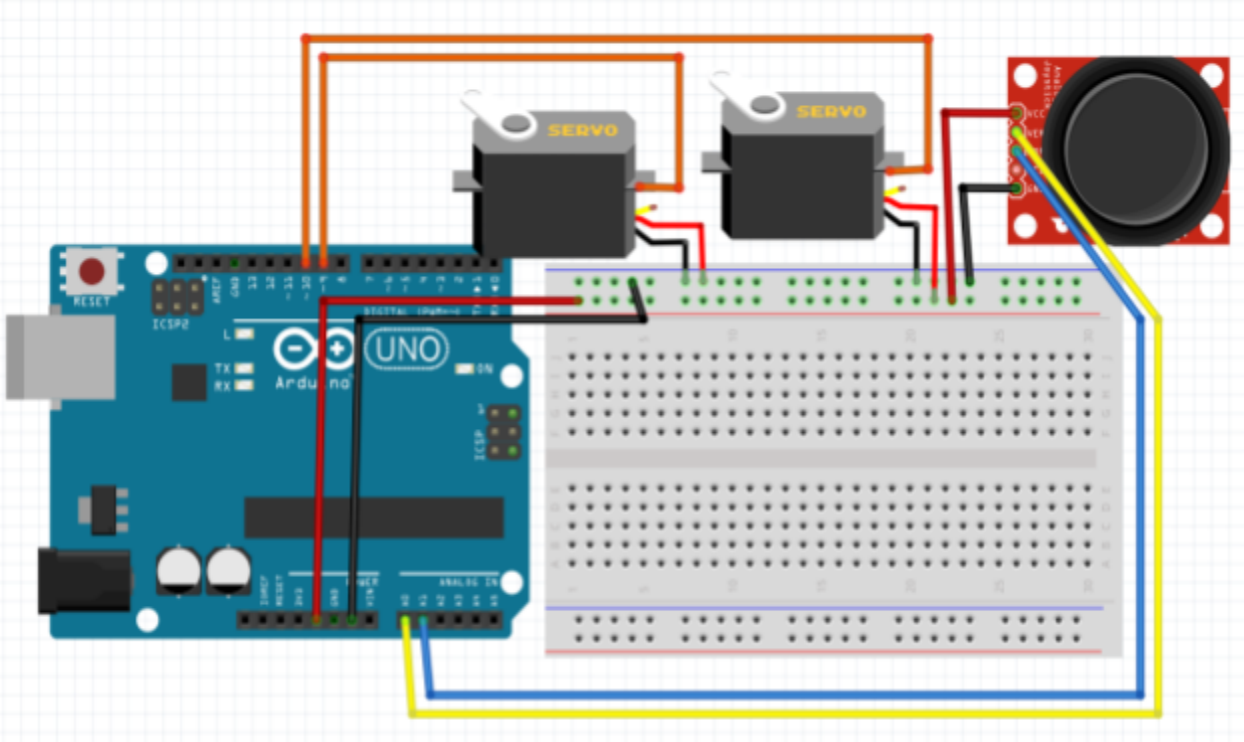


There are 5 connections to the Joystick.
GND = Ground
VRx = X axis (moving the joystick side to side)
VRy = Y axis (moving the joystick up and down)
SW = switch (acts a button) If you do not need a switch, we only use 4 pins.

Analog joysticks basically return a value from 0 - 1024. When the joystick is in the middle, it is returning a value of 512. Therefore when we code using a joystick we need to use a map function to scale these values to degrees that measure between 0 and 180 degrees.

Procedure:

1. Wire the project using the following schematic.



Connecting Joystick to Arduino

- Connect Vrx to pin A1
- Connect Vry to pin A0
- Connect GND to GND via the breadboard
- Connect +5v to 5V via the breadboard

2. **The CODE.** You can create your code in the BlocklyDuino then paste it to the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

- First setup servos. We will need two servo blocks. Assign one servo block to Pin 9 and the other servo to Pin 10.

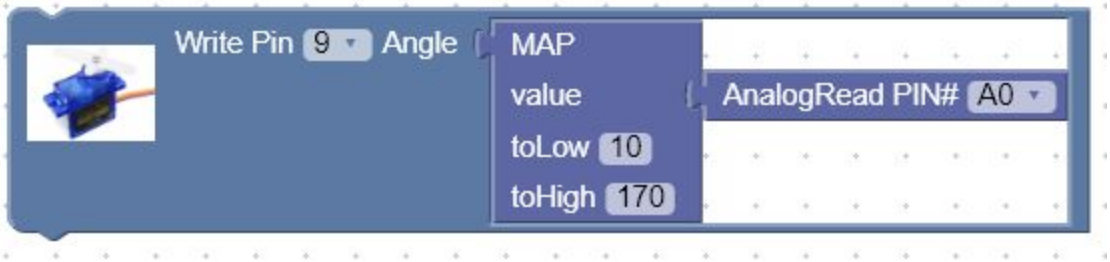


- Remove the Math block from the Servo block and replace it with a map function which is found under Math blocks as shown below

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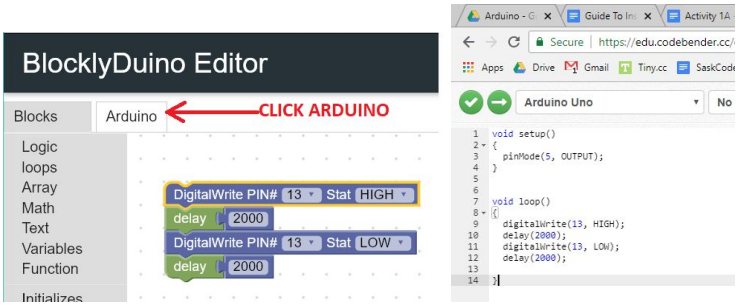


- We now have to connect an analog reading to the map block that will send a value to the servo so that it can rotate from 10 to 170 degrees. That value will depend on the position of the joystick. The joystick x axis is connected to Analog pin A0 and the y axis is connected to Analog pin A1. Remember, you will need to code both servos.



*****SAVE your sketch to your BlocklyDuino Folder that you created*****

4. Copy the Arduino Code from BlocklyDuino and paste into the [Codebender website](). Hook up your arduino to your computer using the USB cable.



5. Upload your program to the arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green “upload” button.

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Assignment: Distance Sensor.

1. Give 2 examples where you would use a joystick and a servo.

2. What are two things you could design using two servos and a joystick?

3. What happens when you move your joystick on an angle? Why does this happen?

Extension Challenges

4. **Change the limits of the servos**: Try editing the values in the map statements so that the movements of the servos change.

5. Using cardboard, tape, or sticky tack to build a basic crane.

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